

ASCII MASTER FLOW – PANEL 1/12: Institutional command map

DEPARTMENT OF SPACE -> policy umbrella and sovereign direction

ISRO -> public R&D, launch vehicles, spacecraft and missions

IN-SPACE -> promote, authorise and supervise non-government activity

NSIL -> commercial contracts, services and technology transfer

RULE -> institution names are not interchangeable

ASCII MASTER FLOW – PANEL 2/12: Centre-to-launch chain

VSSC -> launch-vehicle design and development

LPSC / IPRC -> liquid propulsion and propulsion testing roles

SDSC-SHAR -> integration, range and launch operations

MISSION CONTROL -> flight monitoring and decision chain

RULE -> a centre function is not the whole ISRO mandate

ASCII MASTER FLOW – PANEL 3/12: Policy and law stack

OUTER-SPACE TREATIES -> international State obligations

INDIAN SPACE POLICY 2023 -> executive role allocation

IN-SPACE NORMS -> authorisation and supervision interface

COMMERCIAL CONTRACT -> service and risk allocation

GAP -> policy is not a comprehensive domestic space statute

ASCII MASTER FLOW – PANEL 4/12: Launcher-payload separation

LAUNCH VEHICLE -> supplies ascent energy and guidance

PAYLOAD FAIRING -> protects the mission payload

SEPARATION -> launcher task and spacecraft task diverge

SPACECRAFT -> carries payload and may raise or maintain orbit

RULE -> rocket success and spacecraft service entry are separate

ASCII MASTER FLOW – PANEL 5/12: Orbit vocabulary matrix

LEO -> altitude class, often used by crew and Earth observation

POLAR -> high-inclination ground-track geometry

SUN-SYNCHRONOUS -> repeated local-solar-time crossing

GSO -> orbital period tied to Earth's rotation

GEO -> circular equatorial geosynchronous special case

ASCII MASTER FLOW – PANEL 6/12: GTO-to-GEO rail

LAUNCHER ASCENT -> reaches transfer conditions
GTO INJECTION -> elliptical intermediate orbit
SPACECRAFT PROPULSION -> raises perigee and circularises
GEO SLOT -> circular equatorial final service orbit
TRAP -> GTO delivery is not direct GEO placement

ASCII MASTER FLOW – PANEL 7/12: Operational launcher matrix

PSLV -> versatile workhorse and mixed mission profiles
GSLV -> cryogenic upper stage and transfer-orbit role
LVM3 -> heavier operational class and human-rating base
SSLV -> separate small-satellite launch architecture
RULE -> role, propulsion, orbit and status must travel together

ASCII MASTER FLOW – PANEL 8/12: Lineage and status ladder

SLV-3 -> experimental orbital-access learning

ASLV -> intermediate technology-learning step

PSLV / GSLV / LVM3 -> operational families in owner

SSLV -> developed small-launch family with dated status needed

NGLV -> approved future development, not an operational launcher

ASCII MASTER FLOW – PANEL 9/12: Propulsion comparison

SOLID -> stored propellant and limited throttle control
EARTH-STORABLE LIQUID -> ambient-temperature liquid pair
CRYOGENIC -> liquid oxygen plus liquid hydrogen
SEMI-CRYOGENIC -> liquid oxygen plus kerosene
RULE -> propellant class does not prove flight status

ASCII MASTER FLOW – PANEL 10/12: Human-rating gate

RELIABLE LAUNCHER -> necessary but insufficient base

MARGINS AND REDUNDANCY -> tolerate selected failures

HEALTH MONITORING -> detect unsafe conditions

ABORT CAPABILITY -> protect crew across ascent phases

CERTIFICATION -> separate human-rated configuration

MUST REMEMBER: India's space architecture separates the Department of Space, ISRO...

ASCII MASTER FLOW – PANEL 11/12: Access-to-space system

VEHICLE -> propulsion, structure, avionics and guidance

FACTORY -> repeatable stage and subsystem quality

PAD / RANGE -> integration, safety and launch window

MISSION ASSURANCE -> verification and anomaly closure

OUTPUT -> cadence and customer confidence

CLOSE DISTINCTION: Organisation is not operation, launcher is not spacecraft, stage is...

ASCII MASTER FLOW – PANEL 12/12: Future-claim firewall

DEFINE -> institution, vehicle, spacecraft, orbit and service

CLASSIFY -> retired, operational, approved or under construction

TRACE -> propulsion, infrastructure, integration and assurance

SEPARATE -> technology transfer from authorisation and safety

VERIFY -> dated official capacity, schedule and outcome source

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date every mission and capability claim;...